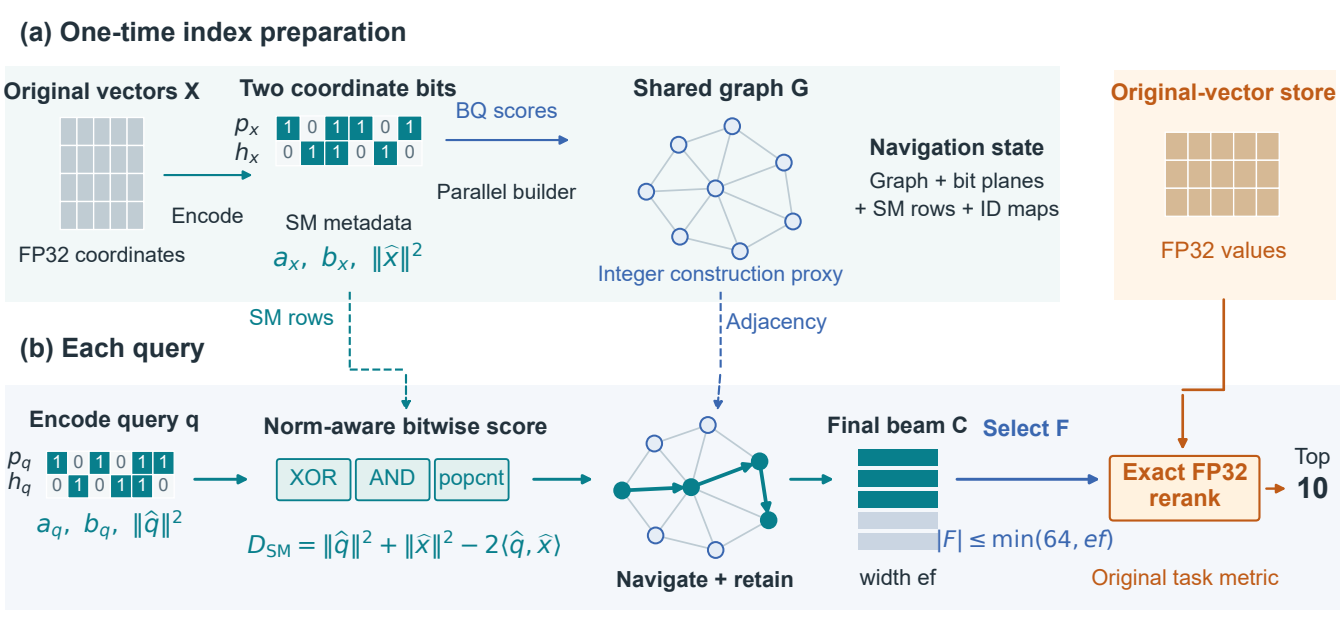

(a) One-time index preparation
Original vectors X
FP32 coordinates
Encode
Two coordinate bits
p_x 1 0 1 1 0 1
h_x 0 1 1 0 1 0
SM metadata
$a_x,\ b_x,\ \|\hat{x}\|^2$
BQ scores
Parallel builder
Shared graph G
Integer construction proxy
Navigation state
Graph + bit planes
+ SM rows + ID maps
Original-vector store
FP32 values
SM rows
Adjacency
(b) Each query
Encode query q
p_q 1 0 1 0 1 1
h_q 0 1 0 1 1 0
$a_q,\ b_q,\ \|\hat{q}\|^2$
Norm-aware bitwise score
XOR
AND
popcnt
$D_{SM} = \|\hat{q}\|^2 + \|\hat{x}\|^2 - 2\langle\hat{q}, \hat{x}\rangle$
Navigate + retain
Final beam C
Select F
width ef
$|F| \le \min(64, ef)$
Exact FP32 rerank
Top 10
Original task metric